

ENGINE LEVEL — IS THE GEOMETRY RIGHT?

Line-of-sight engine

profile sampling · curvature + refraction

5 switchable defects, injected one at a time



Benchmark

15 terrain cases, expectations derived
(critical distance, not intuition)
4 behavioural properties on random terrain



Mutation analysis

run the whole benchmark against
each deliberately broken engine



4 defects killed · 1 equivalent

equivalence proven by paired sweep,
not assumed



DESIGN LEVEL — IS THE INFERENCE RIGHT?

Synthetic landscape

anisotropic ridges · lake flooded to a constant



Ground truth by construction

S0 exchangeable → p exactly uniform
S1 best of K placements → known power
(theoretical ceiling 85 %)



Paired contrasts

correct · water unmasked · tight crop
same data; only the defect differs



Calibration · power · mechanism

what each defect does to inference,
measured against an exact expectation



Diagnostics for any real study

D1: null placements touching water = 0

D2: orientation coverage = 100 %



Only a validated engine and a diagnosed design touch real terrain